

Unit 13, Lesson 3: Determining Points on a Circle



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.



Learning Targets

- ☐ I can determine whether a point is on a circle, inside a circle, or outside a circle.
- ☐ I can find the center of a circle when given two points.
- ☐ I can write the equation of a circle given the endpoints of a diameter.



13.3.1 Warm-Up: Notice and Wonder

Castlerigg Stone Circle is one of the earliest stone circles to be found in Britain and is important in terms of megalithic astronomy and geometry. It is composed of 38 free standing stones, some measuring up to 10 feet high and dating back to the Neolithic period, 4000 to 5000 years ago.



1. What do you notice?

2. What do you wonder?



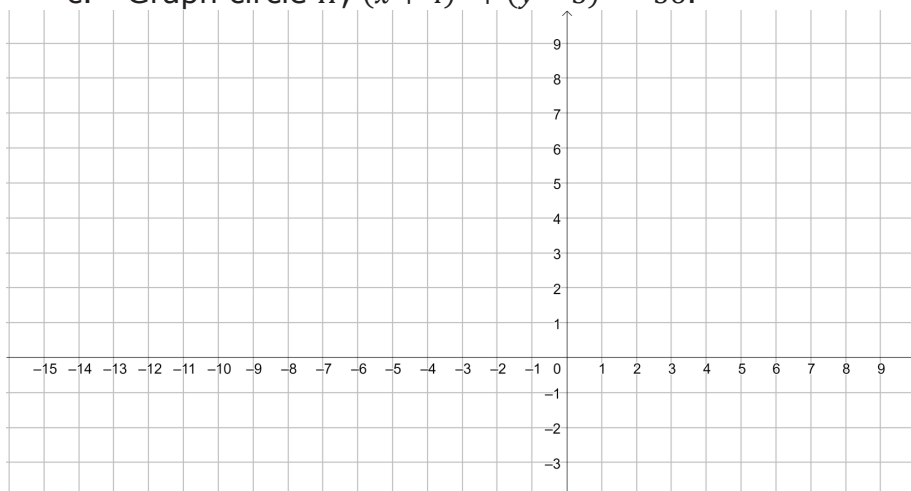
13.3.2 Exploration: Points on a Circle

1. The equation of circle H is $(x + 4)^2 + (y - 3)^2 = 36$.
- a. Complete the table by substituting the given coordinates for x and y in the equation of circle H . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$M(-8, -2)$		$\underline{\hspace{2cm}} = 36$
$B(-3, -1)$		$\underline{\hspace{2cm}} = 36$
$R(-4, -3)$		$\underline{\hspace{2cm}} = 36$

- b. What do you notice about the simplified equations?

- c. Graph circle H , $(x + 4)^2 + (y - 3)^2 = 36$.



- d. Plot points $M(-8, -2)$, $B(-3, -1)$, and $R(-4, -3)$.
- e. What do you notice about the relationship between the location of the points and their simplified equations?
- f. Write a conjecture about the location of a point and the equation of the circle.

- g. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjecture. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Conjecture	Initials

- h. Review your conjecture with your partner. If necessary, make any changes.

2. The equation of circle K is $(x + 6)^2 + (y + 5)^2 = 25$ with points $N(-6,0)$, $V(-3,-1)$, $T(-1,-5)$, and $R(-9,-9)$ on the circle.

- a. What is the length of the diameter of circle K ?

- b. Complete the table by finding the distance between the given points.

Points	Work	Distance
$N(-6,0)$ and $R(-9,-9)$		
$V(-3,-1)$ and $R(-9,-9)$		
$T(-1,-5)$ and $R(-9,-9)$		

- c. What do you notice about the distances between the points?

- d. What can you conclude about points V and R ?

3. Circle W has a radius of $\sqrt{29}$ and points $S(2, -2)$, $D(5, 1)$, and $H(12, -6)$.

a. Determine which two points are the endpoints of a diameter of circle W . Show your work.

b. Describe how to find the center of the circle using the two endpoints of the diameter.

c. What is the center of the circle?

d. Write the equation for circle W .

4. The points $G(3, -1)$ and $H(3, 3)$ are the endpoints of a diameter of circle M . Write the equation of circle M .





13.3.3 Try It: Points on a Circle

1. The equation of circle P is $(x - 8)^2 + (y - 2)^2 = 13$.

Determine whether the following points lie inside, outside, or on circle P .

Point	Work	Location
$(5, -1)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(6, 5)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(12, 2)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(11, 0)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P
$(9, 4)$		☐ Inside circle P ☐ Outside circle P ☐ On circle P

2. Circle T and circle R intersect at point H .

Equation of circle T is $x^2 + (y - 7)^2 = 20$

Equation of circle R is $(x - 7)^2 + (y - 3)^2 = 25$

- a. Which coordinate pair is the location of point H ?
- ☐ $(-4, 5)$
 - ☐ $(7, 8)$
 - ☐ $(2, 3)$
 - ☐ $(4, 9)$

3. Circle S has a radius of 10 and points $F(-6, -6)$, $C(8, 8)$, $W(10, 2)$, and $M(-8, -4)$ on the circle.

- a. Determine which two of the given points are endpoints of a diameter of circle S .
- b. Find the center of circle S .
- c. Write the equation of circle S .



13.3.4 Wrap-Up: Ticket Out the Door

1. The equation of circle W is $x^2 + (y + 4)^2 = 26$.

Select all the points that lie on circle W .

- ☐ (3,0)
- ☐ (5, -3)
- ☐ (1,1)
- ☐ (-3, -2)
- ☐ (-5, -5)

Unit 13, Lesson 4: Partitioning Line Segments



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to partition a line segment.